

James Mwaura

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Electrical Engineering Student

Electrical Engineering student at Seattle Pacific University with hands-on experience in circuit design, embedded systems, and power system analysis. Experienced in debugging hardware, simulating electrical systems, and implementing microcontroller-based solutions. Seeking a Hardware Development Engineering internship

Technical Skills

Programming: C++, Java, Micro Python

Tools: MATLAB, PSCAD, LTspice, Multisim

Hardware: Circuit design, breadboarding, schematic debugging, sensor integration

Embedded Systems: Microcontrollers, GPIO interfacing, real-time signal processing

Analysis: Load flow Analysis: fault analysis, system modeling

Projects

Embedded Ultrasonic Distance Measurement System

Developed microcontroller-based system using ultrasonic sensor and LEDs for real-time distance detection. Implemented signal timing logic to measure distance and provide visual feedback. Debugged inconsistent sensor readings through iterative testing.

Power System Analysis & Fault Study — PSCAD, MATLAB

- Performed load flow analysis using Gauss-Seidel and Newton-Raphson methods
- Simulated fault conditions and evaluated system stability
- Analyzed voltage behavior and system performance under varying load conditions

Active Bandpass Filter Design — MATLAB, Multisim

Designed Sallen-Key bandpass filter and simulated performance in Multisim. Verified frequency response and adjusted component values to meet specifications.

Education

B.S. Electrical Engineering, Seattle Pacific University (Expected June 2027)

Work Experience

Shift Manager — Domino's

Crew Member — Subway